

• Nanoscale Device Engineering (EC 502)



Dr. Manish Kumar Rai
Assistant Professor
ECE Department
IIIT Surat

Syllabus

Unit – I

L = 3, T = 0, P = 0, C = 3

- Introduction to Nanoscience and Nanotechnologies

Unit – 2

- Fabrication Technology & SPICE Device Models

(VLSI Processes: Introduction, Twin-Well CMOS Process, Integrated Devices, MOSFETs, Resistors, Capacitors, pn Junction Diodes, BiCMOS Process, Lateral pnp Transistor, p-Base and Pinched-Base, Resistors, SiGe BiCMOS Process, VLSI Layout, Beyond 20nm Technology Implications of Technology Scaling: Issues in Deep-Submicron Design: Silicon Area, Scaling Implications, Velocity Saturation, Subthreshold Conduction, PVT Variations, Wiring: The Interconnect, SPICE Device Models: The Diode Model, The Zener Diode Model, MOSFET Models, The BJT Model)

Unit – 3

- Applications of Nanoscience and Nanotechnologies

(Information and Communication Technologies: Integrated Circuits, Data Storage, Photonics, Displays, Information storage devices, Wireless sensing and communication)

Data storage devices

- To store data in electronic devices different types of memory is used.
- Memory devices are used to store both the program and data that the processor uses to perform various operations.

Types of Memory:

1. Primary Memory (Main Memory)

- This memory is directly accessible by the CPU. It's fast but usually limited in size. It has two types:
- **Non-Volatile Memory:** Flash, ROM, EEPROM
- **Volatile Memory:** SRAM, DRAM

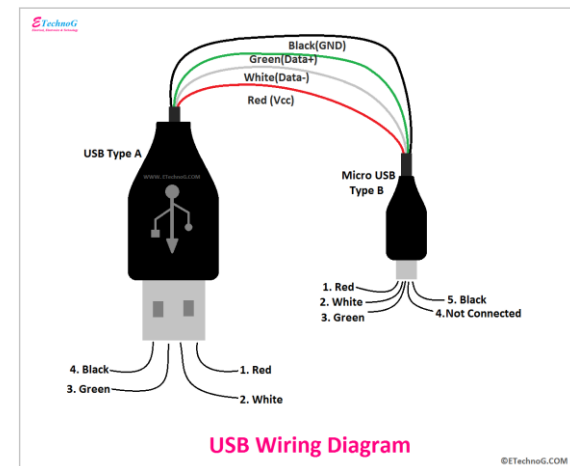
2. Secondary Memory (External or Auxiliary Memory)

- Used for long-term data storage. It's slower than primary memory but has much larger capacity.

Memory in Computer Systems

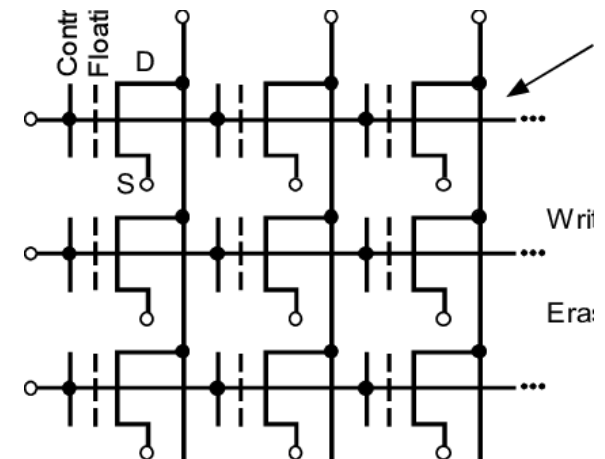
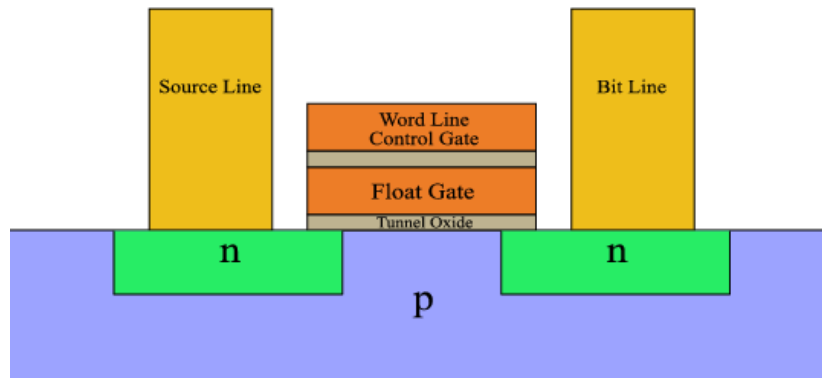
Flash Memory

- Flash memory is a non volatile memory i.e. it retains the data even when there is no power supply.
- This flash memory works on the principle of EEPROM. EEPROM stands for Electrically Erasable Programmable Read-Only Memory.
- Flash Memory can be erased multiple times and update the data or program integrated into it.



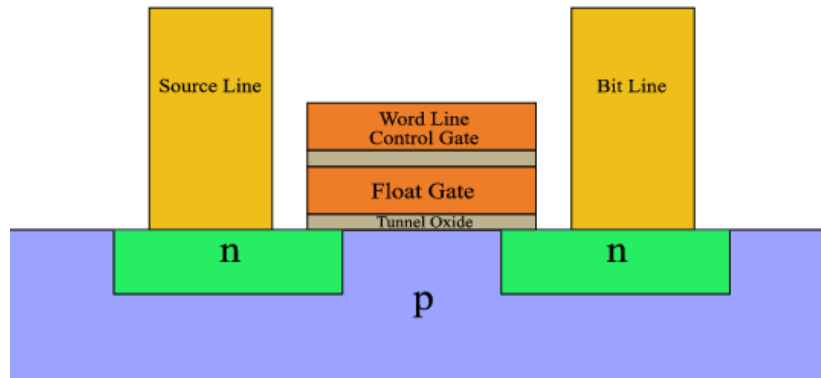
Memory in Computer Systems

- Flash memory is made up of small-small memory cells which are made up of floating-gate transistors.
- All memory cells are organized in the sequential order called an array of memory cells.
- Data can be stored to the hardware in the form of 0 and 1 only. So to store the data into the flash memory we should give the electric supply to add the pattern of 0 and 1 into the flash memory.



Memory in Computer Systems

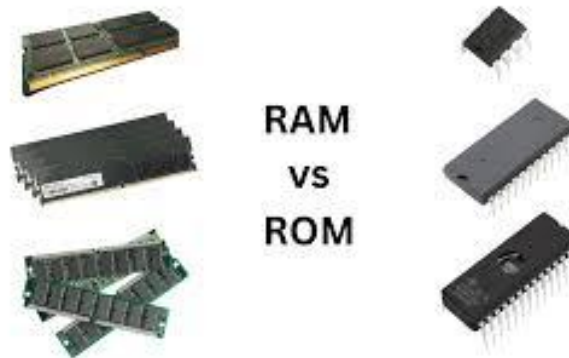
- If no electron is trapped in floating gate then it indicates logic '1' stored. It indicates a conducting channel.
- To read data, we apply $S=0$, $Sub=0$, $CG=3V$, $D=3V$ then current flows between source and drain.
- Logic '0' indicates the trapped electrons in the floating gate. No current is obtained because the trapped electrons have increased the threshold voltage of the MOSFET.
- To write logic '0', we apply $S=0$, $Sub=0$, $CG=10V$, $D=7V$, electrons tunnel through the oxide and trapped in floating gate.
- To erase electrons i.e. to write logic '1', we apply $S=10$, $Sub=10$, $CG=0V$, $D=10V$.



Memory in Computer Systems

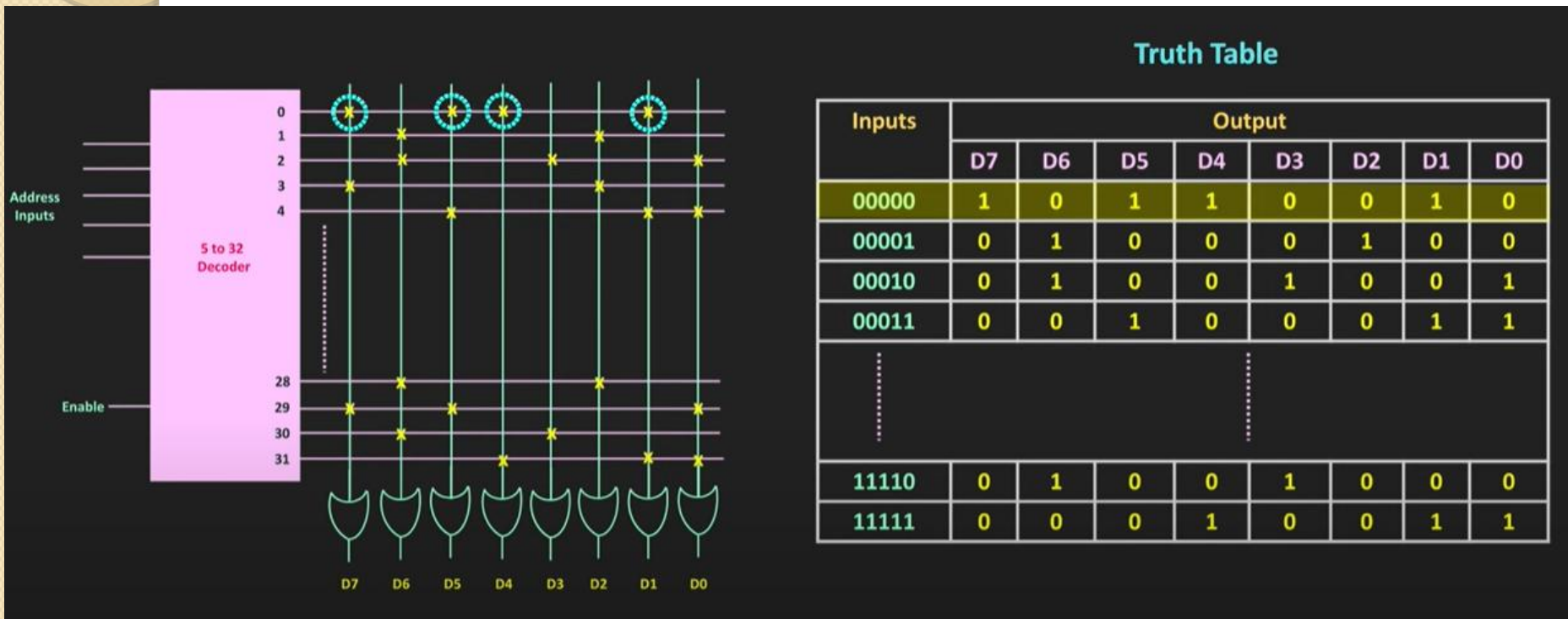
Read-Only Memory (ROM)

- ROM is a non-volatile type of memory that stores permanent data and instructions, like the BIOS, that a device needs to start up and operate.
- Unlike RAM, ROM retains its contents even when power is off.
- While traditionally the data was read-only, modern ROM variants, such as flash memory, can be reprogrammed.



Memory in Computer Systems

Read-Only Memory (ROM)



Memory in Computer Systems

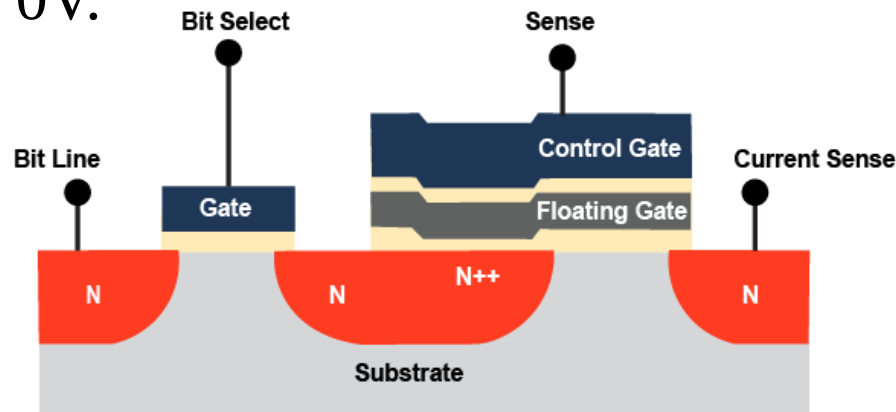
Electrically Erasable Programmable ROM (EEPROM)

- It is a type of non-volatile memory that stores small amounts of data that is retained even when power is off, allowing individual bytes to be erased and reprogrammed electrically.
- EEPROMs are organized as arrays of floating-gate transistors.
- EPROM was being erased using UV light with more erase time (~10-20 min). It was having less number of write/erase cycles (~10k).
- In EPROM part of memory can't be erased.
- EEPROM can be erased electrically in very small time (~5ms) and can have 10^6 write/erase cycles.

Memory in Computer Systems

Electrically Erasable Programmable ROM (EEPROM)

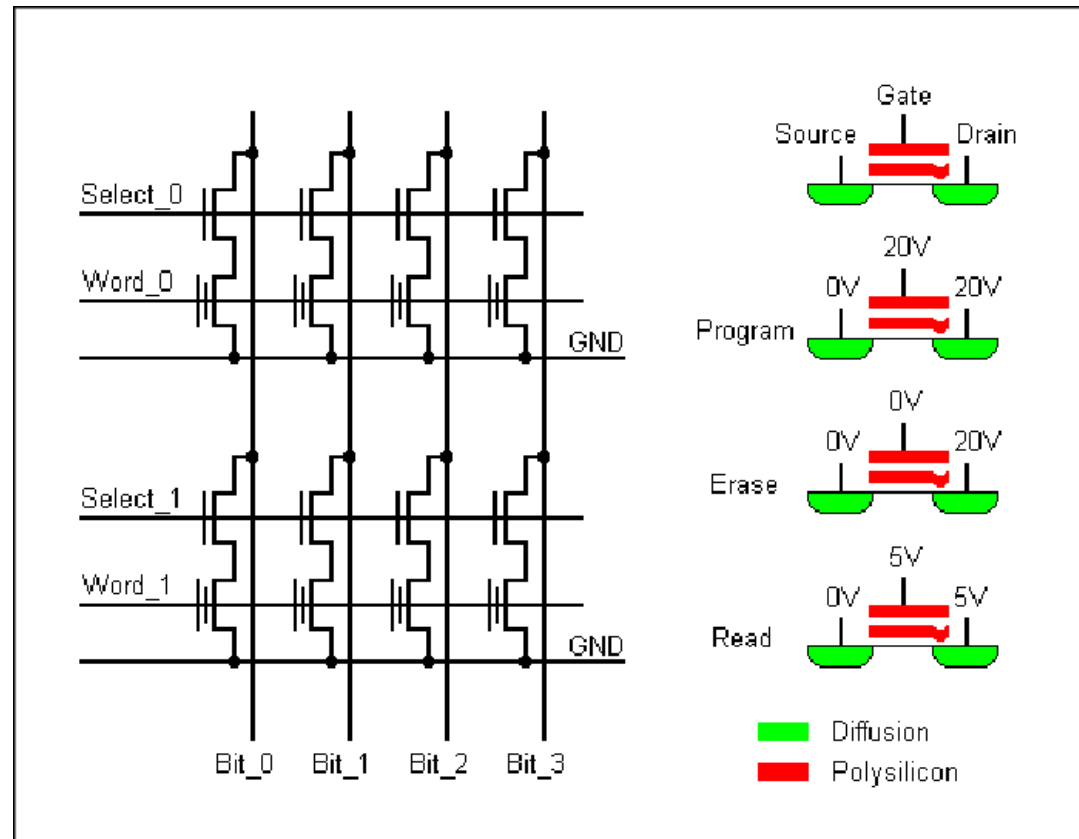
- EEPROM works by trapping electrons on the floating gate of a specialized transistor to store binary data (1s and 0s).
- For higher voltage at gate, electrons move from N^{++} region to the floating Gate and gets stored there.
- To erase it, high voltage is applied to substrate and Gate remains at 0V.



Memory in Computer Systems

Electrically Erasable Programmable ROM (EEPROM)

- Select transistor is used to erase a particular memory block.



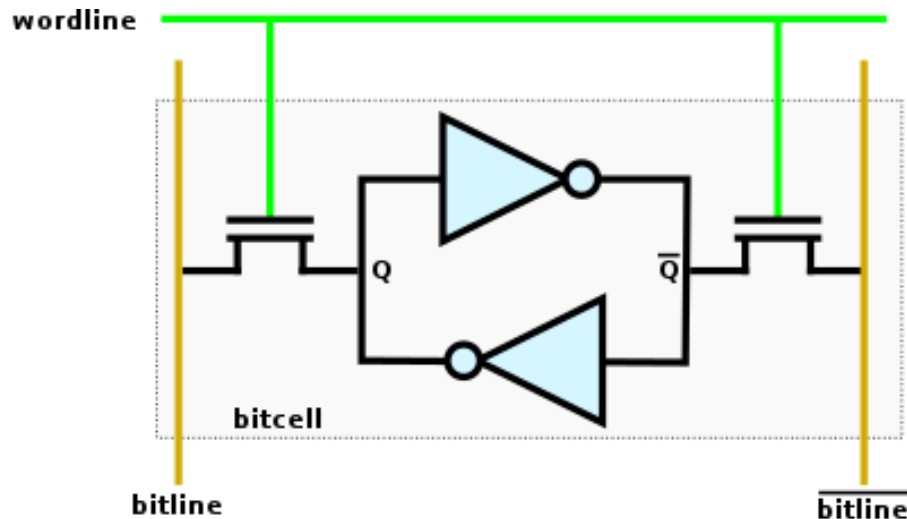
Memory in Computer Systems

Volatile Memory: SRAM, DRAM

- SRAM (Static Random-Access Memory) and DRAM (Dynamic Random-Access Memory) both are volatile memory.
- They lose their data when power is removed, but they differ in how they store data.
- SRAM uses transistors and latching circuits to hold data, making it fast but expensive and used for CPU cache.
- DRAM uses capacitors that leak charge, requiring a refresh operation to maintain data, making it slower but cheaper and used for main system memory.
- It stores the data which is currently being processed by the CPU.

Memory in Computer Systems

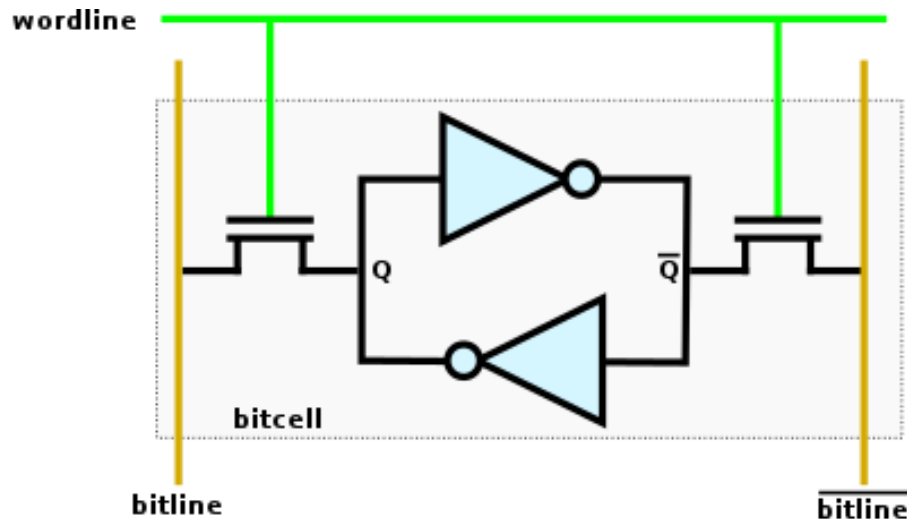
Static Random Access Memory (SRAM)



- **Working**
- **Hold:** wordline = 0, and data is hold in latch mode.
- **Read:** wordline = 1, Access transistors are turned on.
- Bitline and bitline_bar values are read by the amplifier.

Memory in Computer Systems

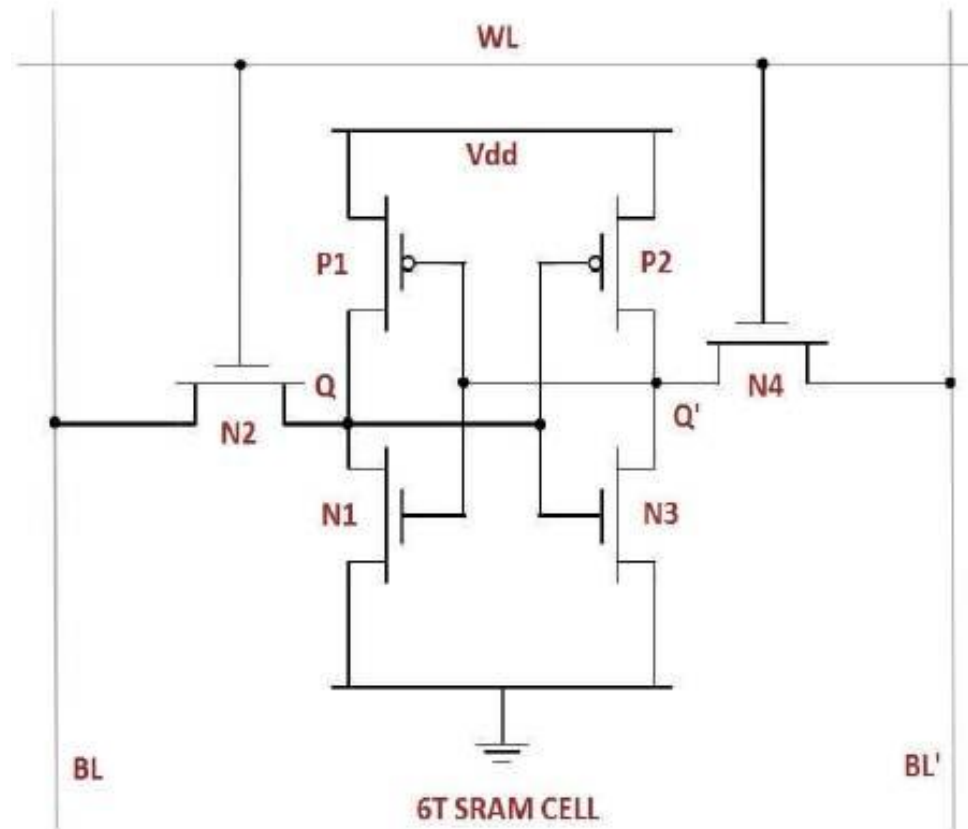
Static Random Access Memory (SRAM)



- **Working**
- Let $Q=0$ and $Q_bar = 1$. wordline = 1, bitline bar = 0.
- Voltage decreases at Q_bar making it 0 and increases at Q making it 1.

Memory in Computer Systems

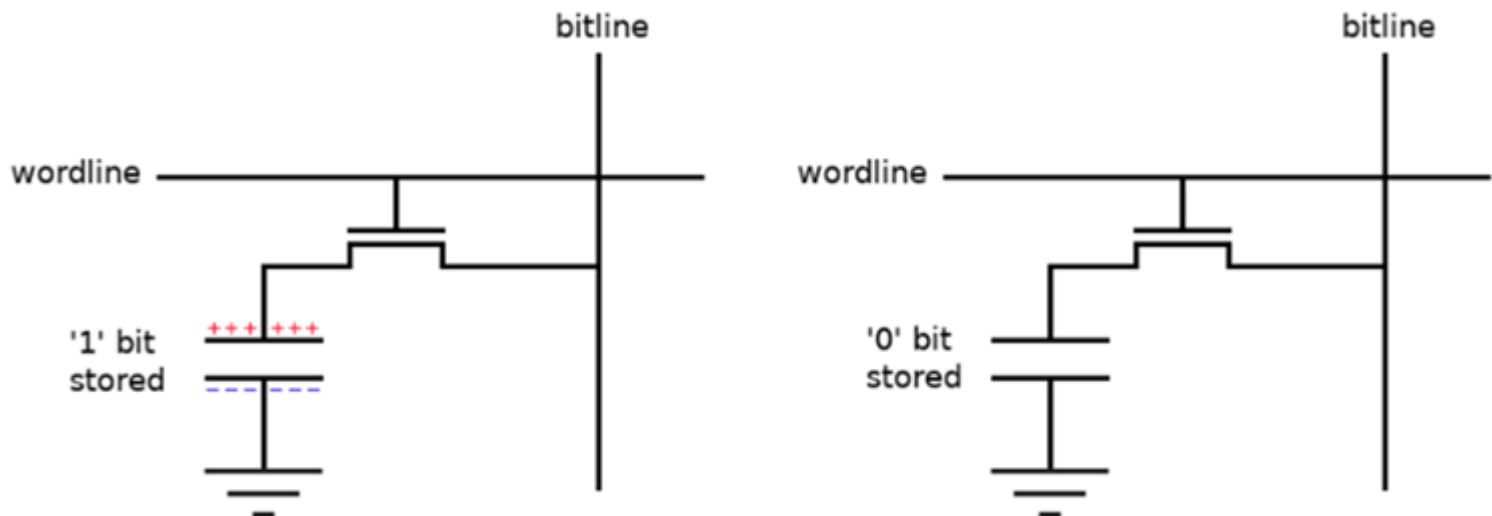
Static Random Access Memory (SRAM)



Memory in Computer Systems

Dynamic Random Access Memory (DRAM)

- The fundamental storage cell within DRAM is composed of two elements: a transistor and a capacitor.
- During a read or write, the wordline goes high and the transistor connects the capacitor to the bitline. Whatever value is on the bitline ('1' or '0') gets stored or retrieved from the capacitor.



Memory in Computer Systems

Dynamic Random Access Memory (DRAM)

- The capacitors leak charge over time therefore, to maintain the data stored in memory the capacitors must be refreshed periodically.

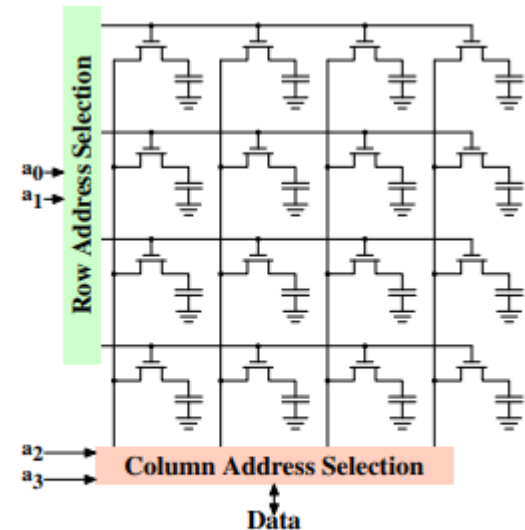
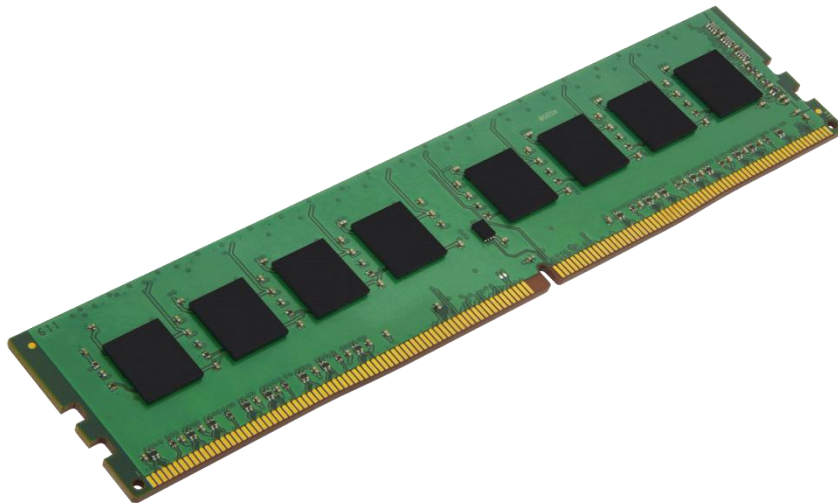


Figure 2.7: Dynamic RAM Schematic

Photonics

- Photonics is the branch of physics that deals with the generation, transmission, detection and control of light (photons).
- It is often considered as the optical counterpart to electronics — where photons play the role that electrons play in electronic systems.
- It is the study and application of light across the spectrum from ultraviolet (UV), through visible, to infrared (IR).
- A photon is a quantum particle of light that carries energy and momentum but has no mass.

Photonics

Major Areas of Photonics

Optical Communication:

- Fiber optic transmission of data using light.

Laser Technology:

- Applications in medicine, manufacturing, and defense.

Optoelectronics:

- Integration of optical and electronic components (e.g., photodiodes, modulators).

Imaging and Displays:

- Cameras, projectors, and optical sensors.

Quantum Photonics:

- Use of single photons for computing and cryptography.

Applications

Telecommunications:

- Internet and data transfer using optical fibers.

Medical Devices:

- Laser surgery, optical coherence tomography (OCT).

Manufacturing:

- Laser cutting, welding, 3D printing.

Defense and Security:

- LIDAR, night vision, optical sensors.

Energyharvesting:

- Solar cells using photonic structures.
- Nanophotonics is a subdomain of photonics that deals with manipulating light at the nanoscale, where classical optics breaks down and quantum effects dominate.

Nanophotonics

Key concepts

- **Subwavelength Control:**
Structures smaller than the wavelength of light can confine and manipulate light beyond the diffraction limit.
- **Plasmonics:**
Involves surface plasmons — oscillations of electrons at metal-dielectric interfaces — enabling strong light confinement and field enhancement.
- **Metamaterials:**
Artificially engineered materials with unique optical properties (e.g., negative refractive index).

Nanophotonics

Key concepts

- **Photonic Crystals:**

Periodic dielectric structures that affect the motion of photons, similar to how semiconductors affect electrons.

- **Quantum Dots and Nanowires:**

Nanoscale light sources or waveguides used in photonic circuits and displays.



Thank You